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Dr. Hashem Mohammadian
Department of Medicine 3
Universitätsklinikum Erlangen
Ulmenweg 18, 91054 Erlangen

Re: Bioinformatics PhD Position — Computational Immunology (Job No. 12900)

Dear Dr. Mohammadian,

I am applying for the Bioinformatics PhD position in Computational Immunology. My interest in this position is not general: the specific pathological focus of your group — autoimmune diseases, mesenchymal activation, inflammatory arthritis, and fibrotic disease — is the subject of theoretical and computational work I have been developing independently over the past year, and the methods your group employs are precisely those needed to test and refine the predictions that work generates.

My paper *On the Thermodynamic Consequences of Categorical Completion on Biological Systems: Specificity through VDJ Recombination Categorical Filter Cascades* proposes that adaptive immunity operates through categorical exclusion rather than molecular complementarity. MHC molecules function as designed ambiguous filters that selectively present peptides from proteins with high categorical richness R — quantifying isoform diversity, conformational entropy, and downstream connectivity — while suppressing peptides from low-R self-proteins. The central empirical result is that MHC binding affinity correlates with parent protein categorical richness independently of classical sequence features: $r = 0.73$, $p < 10^{-12}$, $n = 2{,}847$ IEDB peptides for MHC-I; $r = 0.68$, $p < 10^{-10}$ for MHC-II. Adding the categorical richness term to NetMHCpan improves AUC from 0.68 to 0.81 ($p < 0.001$, DeLong test). The framework predicts autoimmunity as the condition where self-proteins acquire $R > 10^5$ and enter pathogen categorical space. I apply this explicitly to collagen ($R \approx 10^{5.5}$), whose extensive isoform repertoire and dense post-translational modification landscape push it across the threshold — making it a frequent autoimmune target by geometric necessity, not molecular accident. This is the substrate of the pathology your group investigates.

A second paper, *Syndrome: Categorical Resolution of Disease Trajectories*, formalises a computational framework for disease state representation using eight cellular oscillator classes: protein folding, enzymatic catalysis, channel gating, membrane potential, ATP synthesis, genetic expression, calcium signalling, and circadian rhythm. Each class contributes a coherence index $\eta_i \in [0, 1]$, forming a disease vector $\mathbf{D} \in [0, 1]^8$ that encodes the pathological signature across biological scales. For inflammatory arthritis and mesenchymal activation, the relevant components are immediate: elevated D_E captures metalloprotease and cytokine cascade dysregulation; elevated D_{Ca} captures the calcium-dependent fibroblast activation central to synovial pathology; elevated D_G captures the inflammatory gene expression signatures directly measurable by single-cell RNA-sequencing. The framework computes disease trajectories in $\mathcal{O}(N \log N)$ and selects optimal therapeutic intervention targets as $\arg\max_i w_i D_i$ — a sparse linear programme with a computable a priori side-effect bound.

A third paper, *On Disease Epistemology in Blind Receiver: Cellular Ignorance, Group Blindness, and Self-Consistency as the Unique Basis of Disease Detection*, proves that template-based diagnostics are epistemologically impossible at every biological scale and derives the unique viable criterion: loop-holonomy self-consistency of the biochemical circuit. A cell is healthy if and only if the composed constraint map around every reaction cycle returns the identity ($H_\ell = \mathrm{Id}\ \forall \ell$); disease is the condition $H_\ell \neq \mathrm{Id}$ for some ℓ. Numerically, the holonomy test achieves AUC = 1.00 against AUC = 0.896 for template comparison across 25 validation experiments, all of which pass. This offers a fundamentally different diagnostic strategy from current biomarker panels — one that detects circuit-level inconsistency that is locally invisible to individual-node measurement, which is precisely the kind of inconsistency characteristic of chronic autoimmune processes.

The computational infrastructure behind this theoretical work is directly applicable to the experimental modalities your group uses. gospel processes whole-genome VCF files alongside RNA-seq expression matrices at 10^6 variants/second, validated against gnomAD v3.1.2 and UK Biobank; the genetic expression oscillator in the Syndrome framework takes its input from exactly this kind of data. lavoisier is a mass-spectrometry metabolomics pipeline achieving 98.9% feature extraction accuracy against NIST reference libraries, with distributed computing via Ray and PyTorch-based neural networks. helicopter addresses multi-modal microscopy image analysis across fluorescence, brightfield, and spectral modalities — directly applicable to imaging mass cytometry data. I have also built deployable web tools for sequence homology search (vivid-symbolism) and multi-modal biological data observation (hieronymus), demonstrating that I build accessible, reproducible research tools rather than scripts.

My academic background is in computational lipidomics at the Technische Universität München, where I trained in the statistical and multi-omics methods that underpin all of this work. R and Python are both standard tools from that training. I am not proposing to import my framework wholesale into your group's programme — that is not what a doctoral position is for. I am proposing that a candidate who has independently derived and empirically validated a mechanistic theory of why collagen and mesenchymal proteins become autoimmune targets, and built the computational infrastructure to analyse the data your group generates, will contribute productively to that programme from the first week.

I am available from July 2026, fully authorised to work in Germany (Aufenthaltserlaubnis), and would welcome the opportunity to discuss this work with you directly.

Yours sincerely,

Kundai Farai Sachikonye